

Tissue Culture Models of AKI: From Tubule Cells to Human Kidney Organoids

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ABSTRACT

AKI affects approximately 13.3 million people around the world each year, causing CKD and/or mortality. The mammalian kidney cannot generate new nephrons after postnatal renal damage and regenerative therapies for AKI are not available. Human kidney tissue culture systems can complement animal models of AKI and/or address some of their limitations. Donor-derived somatic cells, such as renal tubule epithelial cells or cell lines (RPTEC/hTERT, ciPTEC, HK-2, Nki-2, and CIHP-1), have been used for decades to permit drug toxicity screening and studies into potential AKI mechanisms. However, tubule cell lines do not fully recapitulate tubular epithelial cell properties *in situ* when grown under classic tissue culture conditions. Improving tissue culture models of AKI would increase our understanding of the mechanisms, leading to new therapeutics. Human pluripotent stem cells (hPSCs) can be differentiated into kidney organoids and various renal cell types. Injury to human kidney organoids results in renal cell-type crosstalk and upregulation of kidney injury biomarkers that are difficult to induce in primary tubule cell cultures. However, current protocols produce kidney organoids that are not mature and contain off-target cell types. Promising bioengineering techniques, such as bioprinting and “kidney-on-a-chip” methods, as applied to kidney nephrotoxicity modeling advantages and limitations are discussed. This review explores the mechanisms and detection of AKI in tissue culture, with an emphasis on bioengineered approaches such as human kidney organoid models.

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AKI refers to the rapid decline of renal function, which is diagnosed by increased serum creatinine (sCr) levels. AKI can be the result of various insults, such as drugs, toxins, sepsis, or ischemia-reperfusion injury (IRI), causing renal cell failure.¹ Damage to the major structures of the kidney, including tubule cells, glomeruli, interstitial cells, and vascular cells, leads to the initiation of hemodynamic and inflammatory pathways, resulting in decreased GFR.² Since the concept of toxic nephropathy was

introduced in 1965, clinicians and scientists have sought to prevent, detect, and treat nephrotoxicity.³ Nephrotoxins and nephrotoxins are toxic substances that injure the kidney. Resulting kidney damage includes interstitial nephritis,⁴ glomerulonephritis,⁵ rhabdomyolysis,⁶ crystal nephropathy,⁷ tubular cell toxicity,⁸ and glomerulosclerosis.⁹

In 1987, a renal biopsy study of approximately 100 patients with acute renal failure identified the drugs associated with AKI.¹⁰ This research suggested that

nonsteroidal anti-inflammatory drugs, antibiotics, glafenin, contrast media, diuretics, chemotherapy, and acetaminophen may cause kidney injury.¹⁰ On the basis of the affected nephron segment, the mechanisms of drug-induced nephrotoxicity can be classified into various types.¹¹ The categories include hemodynamically mediated kidney injury, tubular epithelial cell damage, tubulointerstitial disease, glomerular disease, renal vasculitis, thrombosis, and obstructive nephropathy. Most drug-induced nephrotoxicity involves injury to the proximal tubular epithelial cells and podocytes. The epithelial cells in the proximal tubules express specific transporters, so chemicals with an affinity for those transporters can enter the cell and induce damage.¹² The entry of drugs into the proximal tubules is mediated by members of the solute carrier family (SLC), such as organic cation transporter 2 (OCT2/SLC22A2), the organic anion transporters 1 and 3 (OAT1/SLC22A6, OAT3/SLC22A8), and receptors such as megalin and cubulin (Figure 1).^{13–15}

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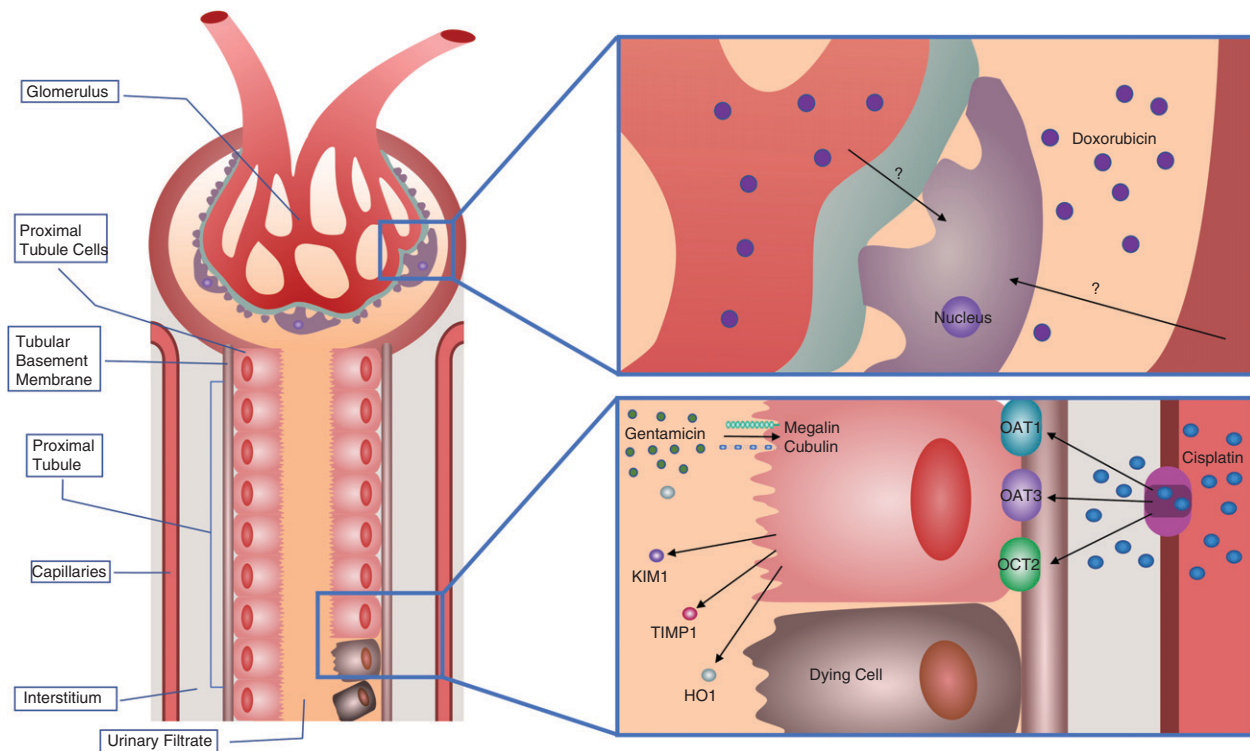


Figure 1. Mechanism of tubular injury during AKI. Nephrotoxicants enter cells through receptor-mediated transport. For cisplatin, transport into the proximal tubule epithelial cell is facilitated by OAT1, OAT3, and OCT2 receptors on the basolateral membrane. Inflammatory and vasoactive mediators and ROS damage the tubular cells and lead to the shedding of the proximal tubule brush border, loss of polarity, and cell death by apoptosis and necrosis. Proximal tubule cell damage releases kidney injury markers, including KIM-1, NGAL, TIMP1, and HO1, into the urinary filtrate. For Dox, the transporters are unknown, but the drug induces oxidative stress resulting in podocyte apoptosis.

In podocytes, the plasma membrane monoamine transporter and bisphosphonates mediate disruption of the podocyte cytoskeleton, causing cell death.^{16,17}

Modeling of kidney injury relies on both *in vivo* and *in vitro* methods. The *in vivo* animal models have been utilized for nephrotoxicity studies for a number of years, with mice and rats being the most popular animals. Meanwhile, *in vitro* models using human kidney cells such as tubule cells, podocytes, or other epithelial cells have also been used to support and further explore findings *in vivo*. In this review, we will focus on the *in vitro* human AKI models. We will discuss biomarkers for nephrotoxicity in humans, advantages and disadvantages of available tissue culture models of AKI, and the latest advances, including human kidney organoids derived from human pluripotent stem cells (hPSCs) and kidney-on-a-chip microfluidic systems.

BIOMARKERS OF AKI

Diagnosis of AKI is on the basis of levels of preclinical or clinical biomarkers. The clinical biomarkers are sCr and BUN, both late-stage markers of AKI. Preclinical studies have identified many early-stage AKI biomarkers, such as enzymes¹⁸ including alanine aminopeptidase, alkaline phosphatase, α -glutathione-S-transferase, γ -glutamyl transpeptidase, and N-acetyl- β -glucosaminidase; proteins including kidney injury molecule-1 (KIM-1),¹⁹ neutrophil gelatinase-associated lipocalin (NGAL), and liver-type fatty acid-binding protein; and cytokines including IL-18. Recently discovered AKI biomarkers include heme oxygenase-1 (HO1) and urinary miRNA.²⁰ Translation of more sensitive markers into the clinic may facilitate earlier diagnosis to implement testing of therapeutics for earlier and more successful treatment of AKI.

KIM-1, also known as hepatitis A virus cellular receptor 1 (HAVCR1) or T cell immunoglobulin and mucin domain-containing protein 1, is a candidate urinary biomarker for detecting drug-induced AKI.¹⁹ In the setting of AKI, the glycoprotein KIM-1 acts as a phagocytic receptor on the apical surface of proximal tubule cells, which enables the removal of apoptotic cellular debris. After a metalloproteinase cleavage, the ectodomain of KIM-1 is shed in the urine. KIM-1 has been tested as a biomarker of drug-induced nephrotoxicity in preclinical trials.²¹ In 2002, Han *et al.* reported the presence of KIM-1 protein in the urine of 23 patients with AKI and nine patients with CKD.¹⁹ Compared with conventional biomarkers, such as creatinine, BUN, urinary N-acetyl- β -glucosaminidase, γ -glutamyl transpeptidase, or alkaline phosphatase, urinary KIM-1 is an earlier diagnostic indicator of

AKI.^{19,22} A study by Timmeren *et al.* showed that KIM-1 expression in kidney tissues correlated with urinary KIM-1, suggesting potential as an early biomarker.²³

HO1 is a 32 kDa enzyme induced in response to oxidative stress.²⁴ It is a cryoprotective antioxidant present in the proximal tubules that is expressed at higher levels, both in plasma and urine, in patients with AKI.²⁵ Deficiency of HO1 after a toxic insult can promote epithelial–mesenchymal transition (EMT) and lead to AKI to CKD transition.^{26,27} Therefore, sustained expression of HO1 after injury might suggest a reduced risk of CKD. After injury, HO1 is expressed in proximal tubules, glomeruli, and interstitial cells.^{28,29}

NGAL, or lipocalin 2, is a protein belonging to the lipocalin superfamily. In a healthy kidney, tubules endocytose NGAL. During injury, NGAL levels rise in both distal tubules and collecting ducts, causing shedding of NGAL into the urine. NGAL found in the urine is comprised of different molecular forms of NGAL originating from various cell types. The increased translation within epithelial cells, secretion by infiltrating neutrophils, and impaired reabsorption all contribute to elevated, detectable levels during injury.³⁰ Elevated levels of NGAL in the kidneys of patients with AKI were first identified in 2005.³¹ The first clinical study evaluating NGAL as an AKI predictor was performed in children at risk of AKI.³² NGAL is under development as a biomarker for the detection of kidney injury, both *in vivo* and *in vitro*.³³ In addition to AKI, NGAL is a promising marker for CKD.³⁴

ILs are the cytokines expressed by lymphocytes, monocytes, macrophages, endothelial cells, and fibroblasts. After kidney injury, tubular epithelial cells and podocytes produce certain ILs as part of the inflammatory response. For example, IL-8 expressed in proximal tubule cells and intercalated renal cells accumulates in urine in response to AKI.^{35,36}

Urinary miRNAs have also been investigated for detecting AKI. Studies

using miRNome profiling revealed that various urinary miRNAs, including miR-21, miR-200c, miR-423, and miR-4640, were differentially expressed by patients with AKI.³⁷ However, the technique does not provide information on the status of renal cells and is not cell specific.³⁸

PRECLINICAL KIDNEY MODELS

Animal Models and Potential Drawbacks

Animal models have contributed a great deal to our understanding of kidney physiology and pathophysiology. Studies in rats and mice have allowed us to explore the pathogenesis, genetics, and underlying mechanisms of kidney diseases. *In vivo* models of nephrotoxic AKI are usually obtained by the administration of toxic drugs to animals.

Differences in gene composition and expression in the mouse and human genomes must be carefully considered when using mouse models.³⁹ Practical drawbacks of animal models include the management, approvals, and cost of veterinary surgical equipment. Training personnel, genotyping, and intensive specialization of labor that is often required to carry out such animal work is time consuming. Species differences in genetics or physiology may result in unexpected outcomes during first-in-human early-stage clinical trials.^{40,41} The failure of many clinical trials of therapeutic approaches to treat patients with AKI could be due to multiple areas of misalignment between the AKI animal models and patients with AKI. This includes the younger age of animals used compared with older patients, the high toxicant dosage usually used to induce AKI in animals as opposed to lower doses given over a longer time to patients who have cancer,⁴² and physiologic differences between species. Although the animal models do not fully mimic clinical AKI pathology, human *in vitro* studies can complement them to address some of their limitations.

Primary Human Kidney Cells

Primary cells are isolated or harvested from living tissues or organs. Renal primary cells permit studies of renal function and the effects of nephrotoxics. Human primary renal proximal tubular cells (HPTC) express proximal tubule cell-specific markers during early passages. These cells express various transporters⁴³ and produce drug metabolism enzymes.^{44,45} However, disadvantages of their use include donor variability,⁴⁶ expression of some transporters and enzymes below physiologic levels, and loss of tubule cell identity during passaging.⁴⁷ All approaches relying on primary human kidney cells have production limitations, including sourcing from discarded kidneys that are unsuitable for transplantation and limited proliferative capacity in tissue culture.⁴⁸

Other Adult-derived Kidney Cells: Advanced Culture and Reprogramming Strategies

Similar to primary cells, kidney organoids derived from kidney biopsy samples can also be used to model nephrotoxicity.⁴⁹ Such human kidney-derived organoids may come from an epithelial progenitor cell type without a stromal component and express a mixed set of markers including aquaporin-1, aquaporin-3, podocin, synaptopodin, and nephrin.

Interestingly, Kaminski *et al.* reprogrammed mouse fibroblast cells with transcription factors *Emx2*, *Hnf1b*, *Hnf4a*, and *Pax8* into early phenotype renal tubular epithelial cells.⁵⁰ Translation of this study to human cells could provide an alternative option to generate tubule cells from other kinds of primary cell types in the future. In a separate direct reprogramming study, human nephron progenitors were generated from adult primary renal tubular epithelial cells by the expression of *SNAI2*, *SIX1*, and *EYA1*, which under certain conditions could be differentiated back into tubular epithelium.⁵¹ Further development of direct reprogramming strategies is needed for large-scale production of desired kidney cells for nephrotoxicity testing.⁵²

Immortalized Human Kidney Cells

To overcome the lack of reliability over many passages and heterogeneity associated with primary cells, researchers have immortalized primary cell types to derive kidney cell lines. Commonly used fully immortalized kidney cell lines include HK-2⁵³ and RPTEC/TERT1.⁵⁴ Primary cells have been immortalized with human telomerase reverse transcription,⁵⁵ human papillomavirus 16 E6/E7 genes,⁵⁶ and/or SV40 large T antigen (SV40T).⁵⁷ Conditionally immortalized proximal tubule epithelial cells (ciPTEC),⁵⁷ and podocytes (CIHP-1)⁵⁸ have the advantage of temperature-controlled states of proliferation versus differentiation (Table 1). Conditionally immortalized cell lines expressing temperature-sensitive SV40T proliferate when incubated at the permissive temperature of 33°C when SV40T is active. On turning off the SV40T by incubating the cells at 37°C, these lines are induced to mature.⁵⁷ The cell type characteristics of the cell lines were confirmed by expression of markers OAT1, OAT3, and OCT2 on the basolateral membrane and ABCB1 (P-gp), SLC47A1 (MATE1), SLC47A2 (MATE2), ABCC2 (MRP2), ABCC4 (MRP4), ABCG2 (BCRP), and the endocytosis receptors megalin and cubilin on the apical membrane.⁵⁹

Kidney-on-a-chip Microfluidic Systems

Microelectromechanical systems allow researchers to apply microfluidics in

tissue culture to engineer an “organ-on-a-chip.” “Kidney-on-a-chip” refers to the use of microfluidic devices for growing renal cells, which can be utilized for preclinical drug development and toxicity screening. The first kidney-on-a-chip study by Jang and coworkers used a microfluidic device lined by living human kidney epithelial cells to produce a system that better mimicked the *in vivo* responses to cisplatin toxicity and phosphoglycolate phosphatase efflux transporter activity than conventional culture conditions.⁶⁰ Fabrication of three-dimensional (3D) renal proximal convoluted tubules embedded within an extracellular matrix on perfusion chips also showed an improvement in phenotypic and functional properties as compared with two-dimensional (2D) controls.⁶¹ In 2018, Vriend *et al.* grew immortalized proximal tubule epithelial cells expressing the OAT1 transporter (ciPTEC-OAT1) on chips to allow for high-throughput screening and compatibility with high-content imaging platforms.⁶¹ In 2017, Musah *et al.* cocultured podocytes derived from human-induced pluripotent stem cells (hiPSCs) with a layer of human kidney glomerular endothelium in an organ-on-a-chip microfluidic device.⁶² They reported improved tissue–tissue interface and molecular filtration properties in the “glomerulus-on-a-chip.”⁶² A recent study using human RPTEC cells cultured on a chip reported improved polarized tight junctions, cilia formation in the apical brush borders, and OCT2

transporter expression on the basolateral membrane, as compared with 2D cultures.⁶³

The kidney-on-a-chip method has many advantages including extended life, biocompatibility, higher gas permeability and sensitivity, and low cost. The major advantage of the kidney-on-a-chip is the ability to apply a nephrotoxicant to either the apical or basolateral surface of the epithelium, which is not possible in a hiPSC-derived organoid. Commercially available kidney-on-a-chip devices such as the TissUse device, OrganoPlate, and the Nortis device are utilized for nephrotoxicity testing and pharmacokinetic studies. Further studies in the field may lead to a more advanced AKI-on-a-chip model and/or a treatment option through the development of the implantable bioartificial kidney.⁶⁴

Stem Cell–derived Human Kidney Models

Although immortalized kidney cells are highly utilized because they are generally easy to grow, they also have reduced function and altered morphologic characteristics as compared with matching cell types *in vivo*.⁶⁵ Because of these limitations, although *in vitro* AKI models on the basis of these cell lines may report decreased cell viability or proliferation, the data do not always translate to follow-up studies conducted *in vivo*.⁶⁶ Therefore, another approach under development is to differentiate stem cells toward a renal lineage for AKI

Table 1. *In vitro* models of AKI in immortalized cells

Cell Line	Markers	Function	Reference(s)
RPTEC/TERT1	Aminopeptidase-N, E-cadherin, MRP2, MRP4, OAT4, MDR1, MATE1, OCTN2, OCT3	Response to PTH treatment by increased cAMP levels GGT activity, Megalin/cubilin transport system	168,169
ciPTEC	Aminopeptidase-N, Pgp, AQP1, dpp-IV, MRP2/4, OCT2, OAT1, OAT3, BCRP, MATE1/2, CaSR, megalin/cubilin receptors	ALP activity, albumin endocytosis, sodium-dependent phosphate uptake, OCT2, and P-gp activity, UGT activity	57,170
HK-2	Vimentin, cytokeratin, $\alpha\beta$ 1 integrin, leucine aminopeptidase, fibronectin, MCT1, MDR1, OATP4C1	Gluconeogenesis, and Na ⁺ -dependent glucose uptake, GGT and alkaline phosphatase activity, increased cAMP levels in response to PTH, synthesis and secretion of plasma proteins	171
Nki-2	E-cadherin, CK8/18/19, GGT1, OAT1, OAT4	Stable cell line, potentially valuable in toxicity and drug ????	172
CIHP-1	Synaptopodin, CD2AP, nephrin, podocin, ZO-1, P-cadherin, α , β , γ -catenin	A valuable tool in the study of human glomerular disease	173

PTH, Parathyroid hormone; cAMP, Cyclic adenosine monophosphate; AQP1, aquaporin-1; ALP, Alkaline Phosphatase; UGT, Uridine 5'-diphosphoglucuronosyltransferase; GGT, Gamma glutamyl-transferase.

modeling. Of interest for AKI research, there are now protocols for the creation of kidney cells induced from hPSCs.^{67–70} For example, hPSCs have been differentiated into other renal cell types including ureteric bud cells, podocytes, tubule cells, and renal progenitor cells.^{70–78}

Self-organizing kidney organoids generated from hPSCs contain various renal cell lineages.^{79–84} The kidney houses two progenitor cell types originating from the posterior primitive streak in the blastocyst, the metanephric mesenchyme and ureteric bud. The metanephric mesenchyme is derived from the intermediate mesoderm whereas ureteric bud is derived from the anterior intermediate mesoderm.⁶⁷ A protocol for the induction of mesoderm and intermediate mesoderm was first published in 2013.⁶⁸ A “kidney-in-a-dish” can be obtained from human stem cells differentiated by providing the appropriate signals and factors at the right times to mimic the developing embryonic kidney.^{79–81,83–85} Kidney organoids developed using various protocols are morphologically and transcriptionally more similar to a human fetal kidney than to an adult kidney.⁷⁹

Most kidney organoid protocols rely on activation of canonical Wnt, Activin A signaling to reach the intermediate mesoderm stage, followed by treatment with fibroblast growth factor 9 for nephrogenesis. Some protocols use a pulse of Wnt activation at the nephrogenesis stage to trigger mesenchymal-to-epithelial transition to increase the number of nephrons formed within the aggregate.⁷⁹ Kidney organoids express markers for nephron segments including distal tubules, proximal tubules, early loops of Henle, presumptive collecting ducts, glomeruli, and podocytes, and endothelial and interstitial cells.^{79,80} Transport into proximal tubules within kidney organoids as measured by dextran uptake suggests the absorptive function is at least somewhat present in these immature organoids.⁸⁰

Although kidney organoids contain many renal cell types, they are not without practical drawbacks including high cost, complexity, skill requirement, variable reproducibility, and limitations to large-scale culture (Table 2). To facilitate scale-up, 3D kidney organoids may be grown in bioreactor spinner flasks.⁸⁶ To lower costs, “KnockOut Serum

Replacement” has been substituted for fibroblast growth factor 9⁸⁷ or special swirling media.⁸⁶ Kidney organoid protocols with improved maturity and reproducibility are needed. Development of more mature organoids expressing drug transporters at levels that are closer to those found in the adult human kidney would further increase the utility of organoid models of nephrotoxicity. Three-dimensional bioprinting permits complex structure fabrication with different materials, cell types, and growth factors.⁸⁸ Bioprinting of hPSC-derived kidney organoids created reproducible, higher-quality organoids, suggesting the translational promise of these methods for large-scale production.^{89,90} At the same time, bioprinting approaches are generally not regarded as simple enough to be widely adopted across research laboratories.

Kidney organoids have many advantages over the cell lines, but they have a short lifespan of approximately 30 days and contain immature endothelial cells that are not well organized into a functional vasculature. New approaches suggest the vascularization challenge is addressable. Organoids grown on chick

Table 2. Advantages and disadvantages of the *in vitro* human AKI models

Method	Advantages	Disadvantages	Reference
Primary cell lines	Intact genetic and phenotypic properties Used with real-time imaging techniques	Interdonor variability Short lifespan Unsuitable for transplantation Source issue	43,44,48
Immortalized cell lines	Long lifespan Less variability Easiest to culture	Absence of important transporters Least differentiated option available	53,54,160
Conditionally immortalized cell lines	Long lifespan Less variability Easy to culture Temperature-controlled state of proliferation	Absence of important transporters	57,58,63
Kidney organoids	System enabling more physiologic modeling Multiple cell types are represented so interactions occur Used with real-time imaging techniques Resources available online at gudmap.org	Highly variable results between cell lines, protocols, and batches Long culture time with a short usage span High cost Maturity level is fetal kidney Presence of off-target cell types Nonadult kidney cell types such as nephron progenitors are present	79,80,89,174
Kidney-on-a-chip	Fluid flow and sheer stress encourages differentiation Multicompartment systems for cell type crosstalk Low cell number eases setup	Uses specialized devices Bioengineering knowledge required Low cell number complicates scale-up and extraction	60,175,176

chorioallantoic membrane had vascularization and circulation of chick blood within kidney organoids.⁹¹ Alternatively, kidney organoids grown under flow on 3D-printed millifluidic chips with a controlled wall shear enhanced the vascular network formation in developing kidney organoids.⁹² Although these studies reported improved vascularization within kidney organoids cultured under flow, the resulting vasculature was still not perfusable.^{92,93} Another approach using renal subcapsular transplantation of hPSC-kidney organoids in mice resulted in vasculogenesis and improved glomerular and tubular maturation.⁹⁴ Three-dimensional bioprinting may provide another avenue for improvement of vascularization and therefore oxygenation within human kidney organoids.^{95,96}

Another challenge is the presence of off-target cell types that are not desired such as neuronal, muscle, and melanoma cells, which increase in relative population as the organoids age.^{97,98} Uncommitted mesodermal cells that exist transiently during kidney organoid differentiation are likely to give rise to these undesirable cell types. Inhibition of brain-derived neurotrophic factor and its receptor NTRK2 greatly reduced the number of neurons without inhibiting differentiation of kidney cell types.⁹⁷ Kidney organoids also do not have stromal patterning, a renal pelvis, an immune system, and a lymphatic system. Although nephrons are patterned appropriately from glomerulus to collecting duct, they do not loop, preventing tubular-glomerular crosstalk.

NEPHROTOXICANTS FOR MODELING AKI

Cisplatin is a chemotherapeutic agent used to treat several cancers including testicular, bladder, ovarian, esophageal, breast, and others. Cisplatin is a cationic drug that enters proximal tubule cells *via* cationic OCT2 and OAT transporters. Cellular injury results from nuclear and mitochondrial DNA damage and oxidative stress, with some efflux

occurring *via* multidrug and toxin extrusion (MATE) transporters.^{99,100} Cisplatin mainly injures the S3 segment of the proximal tubule in the medullary region, causing cell death *via* inflammation and generation of reactive oxygen species (ROS).¹⁰¹

The pathways involved in cisplatin-induced cell death within renal tubular epithelial cells are well established (Figure 1). The damaged mitochondrial DNA reduces oxidative phosphorylation, disturbing mitochondrial ROS (mtROS) production, and leading to apoptosis and necrosis.^{102–104} Cisplatin also causes endoplasmic reticulum stress by activating caspase-12, upregulating Grp78/BiP, and activating the caspase cascade of apoptosis.¹⁰⁵ TNF- α may play a central role in the pathogenesis of cisplatin-induced renal injury by activating the cytokine response.¹⁰⁶

Cisplatin-induced AKI processes have been explored both in animal models^{107–109} and conventional *in vitro* models.^{110–112} For cisplatin, rodent models have AKI biomarkers and histology within 72 hours after one injection of 6–30 mg/kg body weight.^{107–109,113} In HK-2 cells, cisplatin-induced apoptosis causes downregulation of BCL2 and activation of CASP3, suggesting that HK-2–based models mimic the apoptosis pathways *in vivo*.¹¹⁴ Treatment of 3D transwells or suspension kidney organoids with a single dose of 5–50 μ M cisplatin for 24 hours induced toxicity, damaging all cellular compartments in the organoids including podocytes, proximal tubule cells, distal tubule cells, and interstitial cells.^{79,83,84,115}

Digby *et al.* reported that 3D kidney organoids treated with a higher dose of 50 μ M cisplatin for 48 hours had upregulation of proximal tubule injury marker HAVCR1 and the inflammatory cytokine C-X-C motif chemokine ligand 8 (CXCL8), and DNA damage and cell death.^{116(preprint),117} This is consistent with the acute toxicity, high mortality, and absence of fibrosis found in animal models at higher doses.^{116(preprint),117} Digby *et al.* also showed that a repeated low-dose

regimen of cisplatin avoided acute cell death and resulted in a milder but cumulative injury phenotype compared with a single high dose in kidney organoids.^{116(preprint)} Such low-dose regimens better mimic the nephrotoxicity found in patients receiving chemotherapy.

Gentamicin

Gentamicin is a highly charged polycationic aminoglycoside antibiotic that is also a potent nephrotoxicant. Gentamicin enters the cells through megalin, a multiligand endocytic receptor located in the apical brush border, and damages the proximal convoluted renal tubule.^{15,118–120} Gentamicin forms ternary complexes with iron ions, and after oxidation, triggers the generation of ROS.¹²¹ Gentamicin accumulates in lysosomes and produces lysosomal phospholipidosis,¹²² leading to the impairment of the phosphatidylinositol cascade.¹²³ Gentamicin exposure causes an early increase in the proapoptotic protein Bax (Figure 2).¹²⁰

Rodents are injected with 80–100 mg/kg of gentamicin per day for ≤ 8 days to induce AKI.^{124–126} Treatment of HK-2 cells with a 500 μ g/ml dose of gentamicin for 24–48 hours resulted in cellular toxicity.¹²⁷ The gentamicin treatment inhibited the lysosomal activity of phospholipases in the proximal tubular epithelial cells and generated ROS in the cells.¹²⁸ Treatment of kidney organoids with 5 mg/ml gentamicin resulted in KIM-1 expression at the luminal surface of lotus tetragonolobus lectin (LTL)–positive tubules, suggesting that gentamicin treatment injured proximal tubules within the organoids.⁸⁴

Doxorubicin/Adriamycin

Doxorubicin (Dox), sold under the brand name Adriamycin, is a chemotherapeutic drug that increases intracellular ROS production in renal tissue, resulting in cell death *via* apoptosis.¹²⁹ Dox suppresses the activity of topoisomerase 2b, which results in the reduction of both p53-based antiapoptotic activity and antioxidant protein expression.¹³⁰

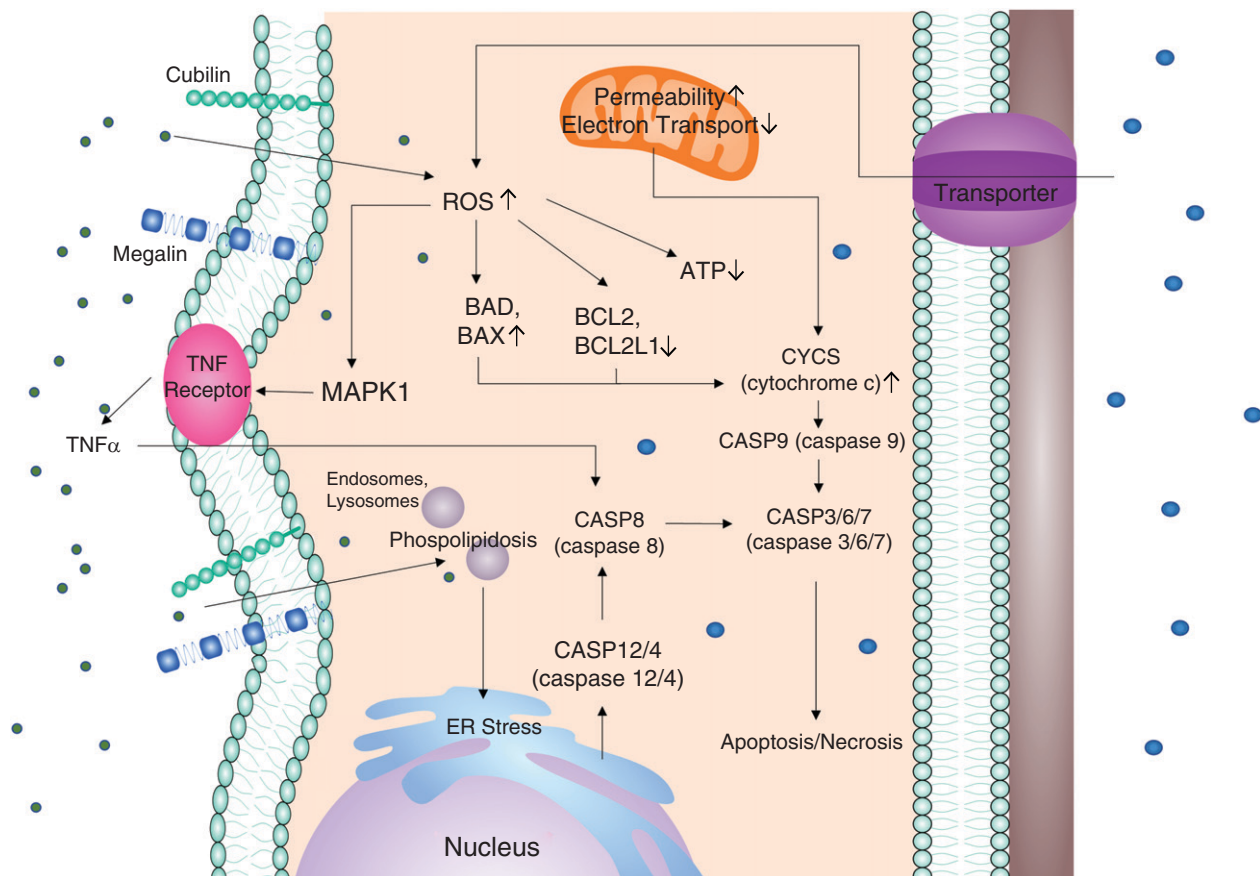


Figure 2. Pathways of nephrotoxicant-induced tubular cell death. Nephrotoxicants enter cells through specific transporters such as OCT2 and OAT1, or endocytic renal receptors such as megalin and cubilin. The drugs damage both nuclear and mitochondrial DNA, leading to the production of ROS and subsequently activation of both mitochondrial and nonmitochondrial pathways of apoptosis and necrosis.^{62,129,167} Mitochondrial dysfunction activates the CASP3 cascade, leading to cell death, whereas the increase of ROS decreases levels of survival proteins such as BCL2 and increases levels of prodeath proteins such as BAX. BCL2, B-cell lymphoma 2; BAX, BCL2 associated X.

Dox-induced oxidative stress disrupts the mitochondrial membrane. Mitochondria then release cytochrome c and proapoptotic factors into the cytosol. Dox upregulates the proapoptotic proteins BAD and BAX and downregulates the antiapoptotic BCL-2 family proteins BCL2 and BCL2L1 (Bcl-XL) in the renal tissue (Figure 2).¹³¹ Studies show that Dox also upregulates BID and caspase-8. Dox mediates cell death through the TNF- α pathway by activating MAPKs, a family of serine/threonine kinases.¹²⁹

Dox is distinct from other nephrotoxicants in that it injures podocytes in the glomerulus¹³² and tubular epithelial cells.¹³³ Injection of 10 mg/kg body weight of Dox has been used to generate a mouse model of podocytopathy.¹³⁴ Kidney organoids have also

been used to create Dox-based podocytopathy models.^{86,89} Kidney organoids treated with 2 μ M or 10 μ M of Dox for 24–72 hours have reduced glomerular structures and tubule networks.^{86,89}

Sodium Cyanide

Cell-culture modeling of hypoxic-anoxic injury is a system parallel to the popular *in vivo* AKI rodent models of IRI.¹³⁵ Anoxia from chemical treatment can model the metabolic stress that results from AKI *in vitro*. ATP depletion via sodium cyanide treatment was first reported in 1996.¹³⁶ Cells were treated with sodium cyanide and 2-deoxy-d-glucose (5–10 mM each) for up to 2 hours to induce stress via tyrosine

phosphorylation of β -catenin, JUP (plakoglobin), and the Bax cofactor NPM1 (nucleophosmin).^{137,138} The cells had ATP content reduced to <10% of normal within 10 minutes. This method has been used for human kidney monolayer cultures including RPTEC¹³⁹ and HK-2140 cells. Implementing the same method to induce anoxia in kidney organoids might further improve this model of AKI to mimic IRI.

QUANTIFYING NEPHROTOXIC INJURY IN VITRO

Kidney-specific *In Vitro* Biomarkers

KIM-1 expression levels and/or staining are often used to assess the extent of cellular damage for AKI studies *in vitro*.

To permit *in vitro* nephrotoxicity screening, luciferase and mCherry transgenes were knocked into the KIM-1 locus *via* CRISPR/Cas to generate a KIM-1 reporter HK-2 cell line.¹⁴¹ The KIM-1 reporter cell line expressed luciferase in response to hypoxia, cisplatin, and glucose treatments under carefully monitored, specific cell-culture conditions.¹⁴¹ However, some *in vitro* studies have found that primary proximal tubule cells and some cell lines did not show significant differences in KIM-1 expression after nephrotoxic drug treatment compared with negative controls.^{33,142} Kidney organoids treated with doses of cisplatin ranging from 5 to 100 μ M have increased expression of KIM-1 (Figure 3).^{83,87} Kidney organoids treated with gentamicin also have upregulation of KIM-1 expression at LTL+ tubules in a dose-dependent manner.⁸⁴ Although kidney organoids upregulate KIM-1, they resemble fetal kidneys, so they may lack the full cadre of transporters involved in drug-induced toxicity.¹¹⁷

An *in vitro* primary proximal tubule cell model of exposure to nephrotoxics revealed a positive correlation between HO1 transcript expression and exposure to nephrotoxic compounds. On the basis of analysis of gene-drug pairs, the *HO1* gene was overexpressed with increasing doses of multiple compounds, including cadmium chloride (CdCl_2), cisplatin, gentamicin, and FK-506.¹⁴³ A kidney-on-a-chip model of exposure to CdCl_2 also upregulated HO1, whereas KIM-1 expression was only marginally increased.¹⁴³ This suggests that HO1 may be a more sensitive readout than KIM-1 when measuring responses to nephrotoxic metals *in vitro*.

Another reported *in vitro* biomarker is tissue inhibitor of metalloproteinase (TIMP-1). Treatment with cisplatin or gentamicin increased the levels of TIMP-1 in animal models.¹⁴⁴ In HK-2 cells, TIMP-1 was significantly increased after exposure to 10 μ M cisplatin.¹¹⁴ Another study in RPTEC/TERT1 cells found that treatment with cisplatin and

gentamicin increased TIMP-1 expression in the treatment groups compared with the vehicle control group.¹⁴⁵ TIMP-1 expression has not been studied in the organoid-based *in vitro* AKI models yet, but may be a useful biomarker for future exploration.

Inflammatory Cytokines

A toxicity model based on HPTC demonstrated an increase in the expression of IL-6 and/or IL-8 at the study endpoint, suggesting that production of these ILs may predict renal proximal tubular toxicity and the potential for preclinical drug development.⁴⁶ Upregulation of IL-6 and IL-8 expression was also found in proximal tubule cell lines in response to gentamicin, CdCl_2 , and uremic toxins. The upregulation of IL-6 and IL-8 was greater than that of KIM-1 in response to these compounds.⁴⁶ Toxic drugs also induced IL-6 and IL-8 in both the HPTC cell line and in hPSC-derived HPTCs.^{146,147}

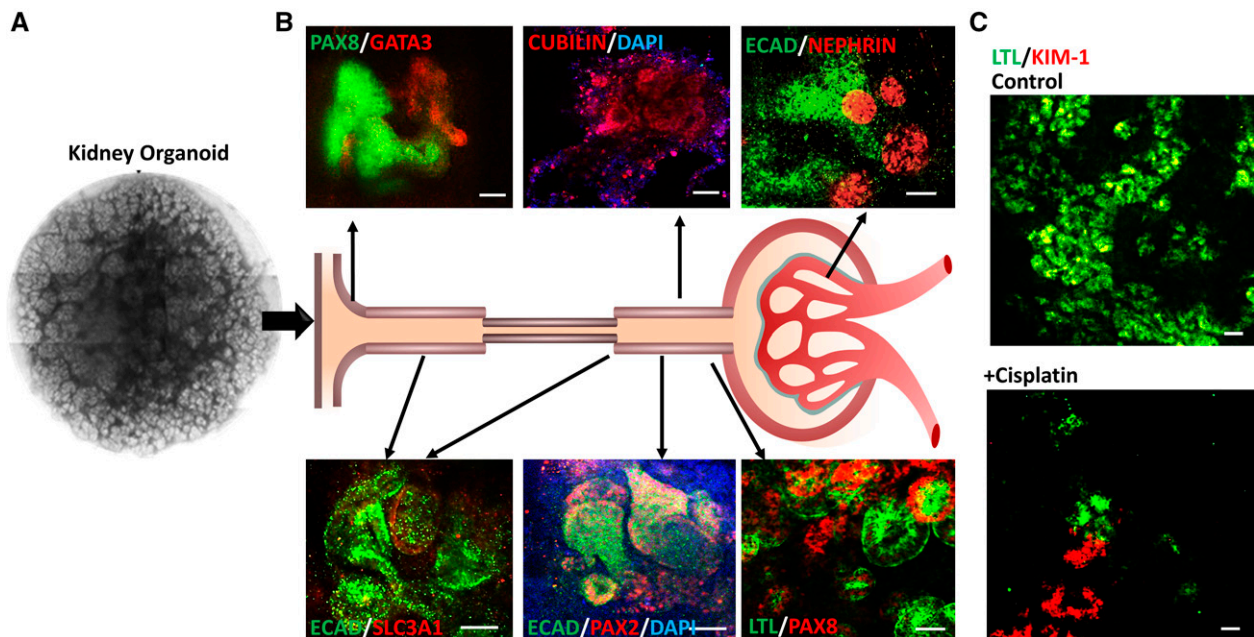


Figure 3. Modeling kidney injury in human kidney organoids. (A) Fully differentiated day 25 kidney organoids created from hiPSCs on transwells. (B) Immunofluorescence of kidney organoids at day 25 shows the presence of various kidney cell types including collecting ducts (GATA3), distal tubules (ECAD), proximal tubules (LTL, PAX8, PAX2, SLC3A1, CUBILIN), and glomeruli (NEPHRIN). Cell nuclei were labeled with DAPI (blue). (C) Representative immunofluorescence of kidney organoid treated with 5 μ M of cisplatin from day 21–23. Cisplatin treatment upregulated expression of kidney injury molecule KIM-1 (red) in the proximal tubule (LTL) compared with the control. GATA3, GATA binding protein 3; ECAD, epithelial cadherin; LTL, lotus tetragonolobus lectin; PAX2, PAX8, paired box homeotic gene 2 and 8. Scale bar 50 μ m.

Cytotoxicity Assays

To evaluate nephrotoxicity *in vitro*, studies commonly rely on assays of acute toxicity, cell viability, cell damage, and/or metabolic activity. Cellular integrity may be measured with dyes such as trypan blue, whereas the metabolic activity may be measured by lactate production or reduction of the tetrazolium dye MTT 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide.^{148–150} Some cytotoxicity assays are not transferrable to 3D kidney organoids, because the compounds do not reach the center of the organoids, but a few studies have explored the possibilities. Apoptosis within the proximal tubule cells can be detected by the presence of cleaved caspase-3. γ H2AX antibody staining and terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) assay may also be used to determine the extent of DNA damage and cell death.¹¹⁷ Colocalization of CC3, γ H2AX, and TUNEL staining together with other markers can provide information regarding the specificity of the toxic drugs for certain cell types or states. CellTiter-Glo luminescent assay can also detect cell viability by measuring bioluminescence activity in renal organoids.⁴⁹

Functional Assays

Several assays are available that test renal cell functionality. Functional assays for permeability include the transport of substrates such as para-aminohippurate and inulin leakage. These have been used for evaluating the nephrotoxicity in animal-based models^{151,152} and for some human-based *in vitro* models.¹⁵³ FITC-labeled BSA transport can also test the functionality of human kidney-derived cells, podocytes,¹⁵⁴ and kidney organoids.^{82,86} Because FITC can enter the cells through solute channels, it may not be a valid functional assay in all patients. Selective endocytosis of dextran, Alexa FluorTM 488 by LTL+ tubule cells has also been used to demonstrate the functionality of organoids.⁷⁹

IN VITRO MODELS OF NEPHROTOXICITY

Conventional *In Vitro* Cell Culture Models of AKI

A common *in vitro* model of AKI utilizes the HK-2 cell line. HK-2 cells treated with 10 μ M cisplatin for 24–48 hours can model some aspects of AKI *in vitro*, including cellular apoptosis and increased kidney injury biomarker expression. HK-2 cells exposed to 20 μ M cisplatin for 48 hours had increased secretion of the inflammatory cytokines IL-6 and IL-8 into the culture media.¹⁵⁵ HK-2 cells treated with 5 μ M Dox also had a significant reduction in cell survival. However, HK-2 cells do not express OAT1, OAT3, OCT2, MRP2, and BCRP,¹⁴¹ and immortalized cell lines express approximately 20-fold more KIM-1 at baseline compared with primary cells, making further induction of KIM-1 expression during *in vitro* nephrotoxicity studies problematic.¹⁵⁶

RPTEC/TERT1 and ciPTEC cell lines express more of the expected tubule cell markers than do HK-2 cells (Table 1). Exposing RPTEC/TERT1 cells to different nephrotoxicants, such as cisplatin, gentamicin, aristolochic acid, or cyclosporine A, induced primary proximal tubule toxicity.¹⁴⁵ RPTEC that were treated with higher doses of cisplatin exhibited a 50%–100% decrease in viability as measured by increased LDH release.¹⁵⁷ Cisplatin treatment resulted in the phosphorylation of p65, increasing the downstream gene expression of IL6, IL8, and C-X-C motif chemokine ligand 10 in RPTECs.¹⁵⁵ The ciPTEC cell line was also used for screening various drugs.¹⁵⁸ Among the different tubule cell lines, ciPTEC cells express the most proximal tubule markers. As discussed above, certain transporters (OAT1, OAT3, and OCT2) and receptors (megalin/cubilin) are important for internalization of drugs. Therefore, ciPTEC cells may be the best model to study human kidney toxicity. Improved toxic drug sensitivity was observed when the ciPTEC cells were grown as 3D structures.^{159,160}

Cultures of immortalized cells on 2D surfaces have the apical surface exposed, so substances cannot enter easily *via* basal transporters. Studies of 3D culture of immortalized cell lines have apical and basal expression and therefore are likely improved.¹⁶¹ Most of these cell lines were derived from the S1 and S2 segments of the proximal tubules in the kidney cortex, so their best application is to S1 and S2 proximal tubule injury.

Immortalized human podocyte cell lines were developed because mature podocytes lose their proliferating capacity and dedifferentiate after isolation. Conditionally immortalized human podocyte cell lines, such as Nki-2 and CIHP-1, are used for modeling podocytopathies *in vitro*.^{58,162,163} A Dox-based *in vitro* study of podocyte injury in conditionally immortalized human podocyte cell lines showed generation of free radicals (ROS).¹⁶⁴ Primary cell line cultures exhibit less sensitivity to drug exposure than do 3D primary organoids.⁶⁶

Advanced *In Vitro* Modeling of AKI in Human Kidney Organoids

hiPSC-derived kidney organoid models have utility for disease modeling and drug screening. The kidney organoids generated are comprised of fetal-stage nephrons expressing both tubule segments, primitive glomeruli, and stromal populations. Patient-derived and CRISPR-mutated kidney organoids derived from hPSCs can recapitulate genetic kidney diseases.^{80,83} Correction of patient hiPSC-derived kidney organoids can restore the function of specific cell types *in vitro*.⁸⁵ Modeling of AKI with nephrotoxicants is facilitated in organoids that have a basal-out pattern that exposes the transporters.

The dosages of drugs required to induce toxicity in kidney cells varies, depending on the method of culture. Between 5–20 μ M cisplatin is required to cause proximal tubular cell apoptosis in organoids cultured using either the transwell or monolayer methods.^{79,84} In our experience, kidney organoids cultured on transwells with 5 μ M cisplatin treatment for 48 hours have upregulated

KIM-1 in the proximal tubules accompanied by loss of expression of the proximal tubule marker LTL (Figure 3). However, cell death in cisplatin-induced AKI organoid studies may not be completely specific to the proximal tubule, because distal tubule cells and interstitial cells are damaged as well.¹¹⁷

Comparative analysis of fetal and adult human kidney transcripts to better understand the lack of proximal tubule specificity was conducted.¹¹⁷ Although matured kidney organoids express SLC31A1/copper transporter 1, SLC31A2/copper transporter 2, ATPase copper transporting beta, and SLC47A1/MATE1, the expression of proximal tubule-specific SLC22A2/OCT2 and SLC47A2/MATE2K transporters was very low.^{116(preprint)} In the case of 3D organoids, a higher 50 μ M dose of cisplatin induced both HAVCR1/KIM-1 and CXCL8 expression and DNA damage and cell death in the organoids.^{116(preprint)} Transcriptional profiling of the corresponding injured organoids showed increased expression of injury biomarkers, such as CXCL2, CCL2, HAVCR1, and NGAL, with high enrichment in TNF- α signaling.^{116(preprint)} The study suggested that cisplatin-induced kidney organoid damage might be mediated through TNF- α receptors (Figure 2).

Human embryonic stem cell–derived kidney organoids treated with 5 mg/ml of gentamicin for 48 hours had KIM-1 expression in LTL+ but not E-cadherin+ tubules.⁸⁴ Dox-induced renal cell death has also been evaluated in kidney organoids derived from hPSCs. Bioprinted 3D kidney organoids treated with 10 μ M Dox for 72 hours exhibited near-complete collapse of glomerular structures and tubule networks.⁸⁹ Organoids treated with Dox did not show podocyte staining for MAFB, so they may have suffered from podocyte loss *via* apoptosis.⁸⁹ Gene-expression analysis found an increase in both BAX and caspase-3 after Dox treatment. Bioprinted 3D kidney organoids have also been used for glomerular toxicity screening.⁸⁹ Dox treatment of these

bioprinted organoids collapsed glomerular structures and reduced their MAFB staining. Kidney micro-organoids treated with Dox had TUNEL positivity in the podocytes, suggesting the feasibility of using these kidney micro-organoids for drug toxicity screening.⁸⁶

Transplantation of human kidney organoids improved both polarization and development of apical brush border,¹⁶⁵ but become an *in vivo* model system without the advantages of an *in vitro* model system. The long-term follow-up study reported chondrogenesis and cyst formation within the grafts, which might be caused by the nonkidney or partially differentiated cells within the transplant.¹⁶⁵ Additionally, there are no studies documenting the effects of nephrotoxic agents on transplanted kidney organoids. Improvements in kidney organoid culture to address these issues would increase the utility of organoid models of AKI.

Kidney-on-a-chip Microfluidic Models of AKI

In 2016, Adler and coworkers reported the use of kidney-on-a-chip to model nephrotoxicity. In the study, HPTCs were successfully self-assembled to recapitulate the *in vivo* structure within the chips. Exposure of HPTCs with 25 μ M CdCl₂ resulted in upregulation of HO1 and KIM-1, suggesting toxicity.¹⁴³ Another nephrotoxicity model used RPTECs-on-a-chip treated with cisplatin, gentamycin, or cyclosporin A. Cell viability was significantly decreased after exposure to the toxicants.¹⁶⁶

In 2017, Musah *et al.* used a microfluidic device to generate “glomerulus-on-a-chip” by seeding primary glomerular endothelial cells in the lower channel and hiPSC-derived podocytes in the upper channel to study the tissue–tissue interaction. They reported that Dox treatment resulted in dose-dependent delamination and decreasing cell viability. Nonselective albumin leakage in the glomerular filtration barrier

enhanced the uptake of albumin by the hiPSC-derived podocytes lining the device’s urine compartment.⁶² A study using human RPTEC cells on chips modeled cisplatin toxicity in cells exposed *via* the basolateral membrane, which was abolished in the presence of the OCT2 inhibitor cimetidine.⁶³ These findings show that advanced models may permit greater understanding of the role of kidney epithelial polarization and membrane localization for drug influx mechanisms, and achieving drug sensitivity at clinically relevant levels.

No AKI model completely mimics clinical AKI, but combining animal models with human tissue culture has proven to be the most robust option available so far. Primary renal tubule epithelial cells and immortalized tubule cell lines have many transporters and are susceptible to injury by nephrotoxic drugs. Compared with immortalized tubule cell lines, tubule cells within kidney organoids derived from hPSC are more sensitive to nephrotoxic injury. However, the younger technologies have their own drawbacks because they are still under development. These limitations include lack of proper maturation and vascularization, the presence of off-target nonrenal cell types in organoid models, and variation between batches. Advancing biomedical engineering approaches including microfluidics, 3D bioprinting, and *in vivo* transplantation may help to mature and vascularize kidney organoids, providing many new avenues for kidney nephrotoxicity modeling.

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